

Lab 2: HTML/CSS & Javascript

animations

This lab session is dedicated to animating objects using HTML5/CSS3 & Javascript (on the client side). We will use our lab 1 Curriculum Vitae (CV) and add animations to it. To begin with, create a directory “TGWK12-lab2” somewhere on your computer. It will be the working directory for lab 2. Copy the files you used during lab 1, your HTML CV, your associated CSS and the image.

1) HTML5/CSS3 animation

Add a CSS3 code to make the image move, rotate, and scale a little on hover with the transition and transform properties in CSS:

```
#photoProfile:hover {  
  transition: transform 3s ease;  
  transform:  
    translatey(100px) /* Translation */  
    rotateY(180deg)   /* Rotation */  
    scale(1.2);       /* Scale */  
}
```

The transition works but, is not nice when you stop the hover. Can you find a way to make it nicer and smoother when quitting the hover?

2) Animations using Javascript (on the client side)

In this exercise, we will use Javascript in the HTML file (client-side Javascript) to animate a new image.

Add an image in your HTML code:

```
<div id="image1">  
  <!-- laptop image from: (svgsilh, 2024, https://svgsilh.com/image/33420.html) -->  
    
</div>
```

Add a script to move the image horizontally to the right:

```
</div>
<script>
  window.addEventListener("load", (event) => {
    const intervalDuration = 1000; // Adjust the interval duration in milliseconds
    let direction = 5; // positive for increasing, negative for decreasing
    var imgObj = null;

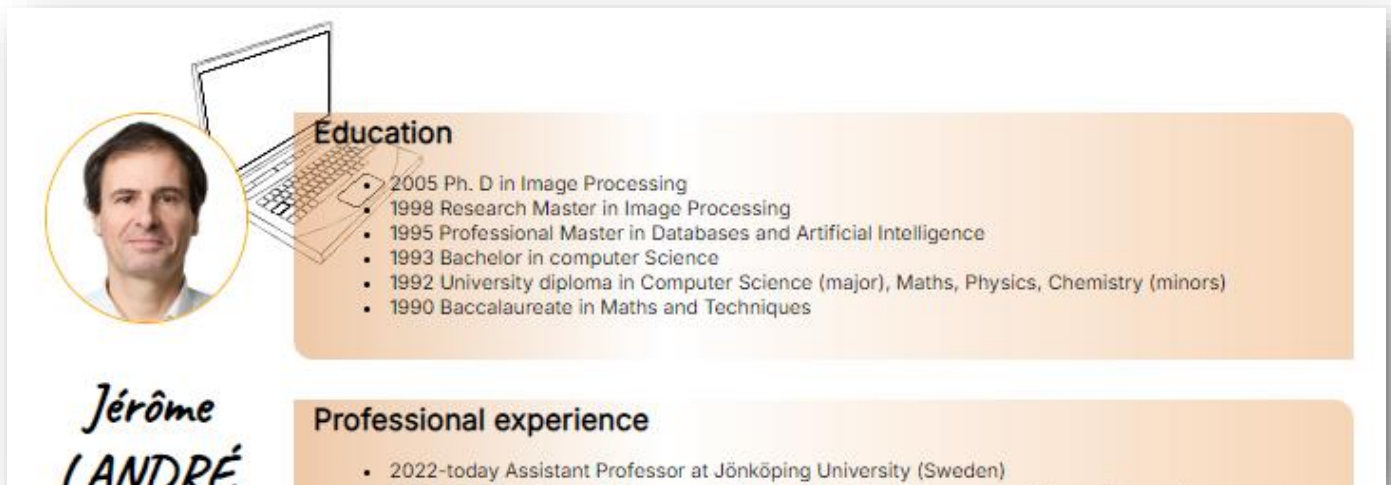
    function init() {
      imgObj = document.querySelector('#image1');
      imgObj.style.position= 'absolute';
      imgObj.style.left = '0px';
      console.log('screen: '+screen.width+' '+screen.height)
      console.log('window: '+window.innerWidth+' '+window.innerHeight)
      const intervalId = setInterval(updateImagePosition, intervalDuration);
    }

    function updateImagePosition() {
      stringLeft=''+imgObj.style.left
      console.log('stringleft: '+stringLeft)
      const substr = stringLeft.substr(0, stringLeft.length-2); // returns 'World'
      console.log('substr: '+substr)

      imgObj.style.left = parseInt(substr) + direction + 'px' // Update the image's left position
    }

    init()
  });
</script>
</body>
</html>
```

You must adapt this code to your own HTML code!



The image shows a CV for Jérôme Landré. It includes a circular portrait of a man, a laptop icon, and two main sections: 'Education' and 'Professional experience'. The 'Education' section lists degrees from 1990 to 2005, and the 'Professional experience' section lists his role as an Assistant Professor from 2022 to the present.

Education
• 2005 Ph. D in Image Processing
• 1998 Research Master in Image Processing
• 1995 Professional Master in Databases and Artificial Intelligence
• 1993 Bachelor in computer Science
• 1992 University diploma in Computer Science (major), Maths, Physics, Chemistry (minors)
• 1990 Baccalaureate in Maths and Techniques

Professional experience
• 2022-today Assistant Professor at Jönköping University (Sweden)

There is a problem when the image reaches the right side of the web page: it disappears because it goes forever to the right...

How can you change the code to make it go backward when touching the right of the page?

Then you will have the symmetric problem: when the image goes to the left and reaches the left border of the page, it will go on to the left forever.

Modify your code to prevent this behavior: we want to go from left to right and then right to left when the image touches the border of the page.

Add a second image (I have chosen a football ball, but you can change it) and do a vertical movement from top to bottom and then from bottom to top. Remember to make the image bounce when it reaches the top and bottom borders of the page.



Jérôme
LANDRÉ

*Teacher &
Researcher*

✉ jerome.landre@ju.se

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Professional experience

- 2022-today Assistant Professor at Jönköping University (Sweden)
- 2006-2022 Assistant Professor at Université de Reims Champagne-Ardenne (France)
- 1995-2006 Software Engineer at Université de Bourgogne (France)

Research

- Image segmentation
- Image comparison
- Object tracking on a video
- Wavelets transform
- Sparse representation

Teaching

- HTML/CSS
- Javascript
- Node.js
- PHP
- Introduction to Image Processing

Computer skills

- HTML5 and CSS3
- Javascript
- Node.js
- Python

Other

- Driving license and personal car
- Football player (less and less)
- Jogger (more and more)

3) Node.js and Express

Test if you have Node.js installed on your computer, open a Terminal and type “node -v”. If you get “22.19.0”, congratulations, you have the good version of node. If not, please install Node version 22.19.0.

When Node is working in your terminal, add a Node.js project (npm init -y) into your directory “TGWK12-lab2”. Add Express (npm install express).

Create a file “server.js” to serve your CV files to a client on port 4321, on route “/”.

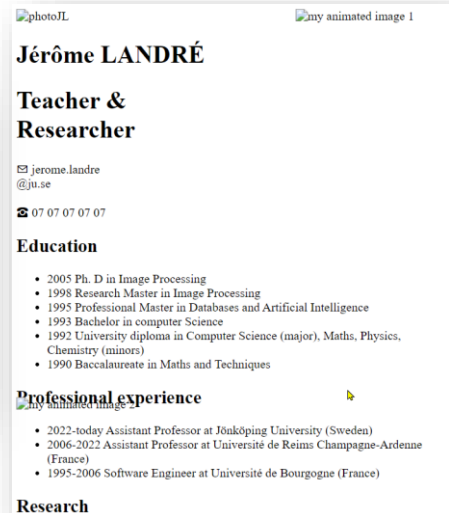
```
const express=require('express')

const port=4321

const app=express()

app.get('/', function (req, res) {
  res.sendFile(__dirname + '/mycv.html')
})

app.listen(port, function () {
  console.log('Server up and running, listening on port '+`${port}`+'...  :')
})
```

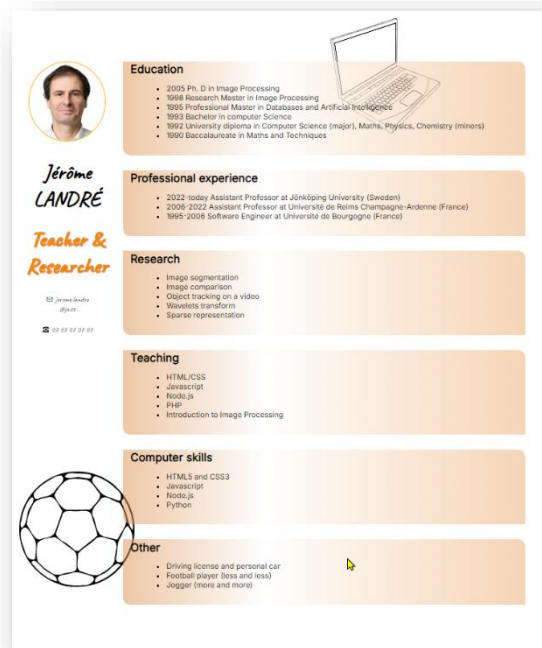


If your style and images do not appear, perhaps you missed declaring a “public” path into your server code...

You must reorganize your code: create a directory ‘views/’ and a directory ‘assets/’ in your lab2 directory. Create a directory ‘assets/css/’, a directory ‘assets/img/’. Move your HTML code to ‘views/’, your CSS file into ‘assets/css/’ and your images to ‘assets/img/’. Modify the paths to these files in the HTML and CSS codes.

Make your assets directory accessible from the server...

```
app.use(express.static('assets'))
```



Re-run until it works! Ask for help if it doesn't work. :)